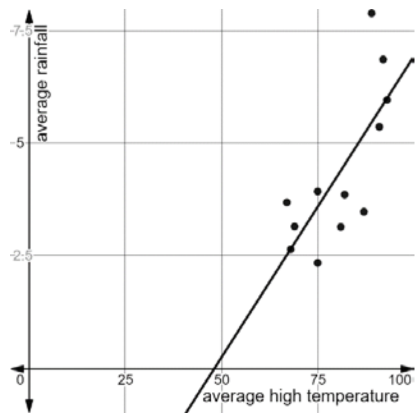
Algebra 1 | Name____Answer key_____
Unit 3 | Date___________________
Lesson 7 Practice | Period_________________

A table shows the average high temperature and the average rainfall each month for a year in Jacksonville, Florida.

Month	Jan	Feb	Mar	Apr	May	Jun	July	Aug	Sept	Oct	Nov	Dec
Average High Temperature (°F)	67	69	75	81	87	91	93	92	89	82	75	68
Average Rainfall (in)	3.69	3.15	3.93	3.14	3.48	5.37	5.97	6.87	7.9	3.86	2.34	2.64

The scatter plot shown represents the data from the table with a linear equation fit to model the equation.



This screenshot shows the slope, y-intercept, and correlation coefficient associated with the linear relationship modeled on the scatter plot.

$$y_1 \sim mx_1 + b$$

STATISTICS

$r^2 = 0.5537$

$r = 0.7441$

RESIDUALS

e_1 plot

PARAMETERS

$m = 0.134323$ $b = -6.48488$

1. Write the equation for the line of best fit shown.

$y = 0.134323x - 6.48488$

2. Interpret the meaning of the slope in terms of the context.

As the average high temperature increases, the average rainfall increases.

3. Interpret the meaning of the y-intercept in terms of the context.

When the high temp is $0°$, the average rainfall is -6.48488 which doesn't make sense in this context.

4. Interpret the meaning of the correlation coefficient in terms of the context.

Strong, Positive, Linear

5. Use an inequality to express an appropriate domain in terms of the context.

$60 \leq x \leq 95$

6. Use an inequality to express an appropriate range in terms of the context.

$0 \leq y \leq 8$

7. Use the model to predict the average rainfall in a month with an average high temperature of $97°\mathrm{F}$.

$y = 0.134323\ (97) - 6.48488$
$y = 6.54$ inches

8. Use the model to predict the average rainfall in a month with an average high temperature of $55°\mathrm{F}$.

$y = 0.134323\ (55) - 6.48488$
$y = 0.90$ inches

9. Use the model to predict the average temperature in a month with an average rainfall of 3 inches.

$3 = 0.134323x - 6.48488$
$x = 70.6°$

10. Use the model to predict the average temperature in a month with an average rainfall of 5 inches.

$5 = 0.134323x - 6.48488$
$x = 85.5°$